

A STUDY OF BRAIN HUMAN INTER FACE USING EEG , MEG

PROSTHETICS LIMBS

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1.INTRODUCTION

Direct connectivity between the human brain and external devices, known as brain-computer interfaces or BCIs, are groundbreaking technologies enabling users to work with technology purely by thought. This ability has a huge potential mainly in healthcare, neuroprosthetics, and assistive technology; it allows people with disabilities to be more freely available in their environment (Maksimenko, 2021).

BCIs enables users to directly hook up through external devices to bypass the conventions channels such as muscles or nerves and offer new means by which it enables severely physically limited individuals to interact with the outer world. Key Concepts in BCIs include: This has included how they can be used to monitor brain health, control robots and enhance cognitive ability. These systems can be reactive, or active, and each offers a different method of interpreting brain activity to help users achieve better results (Hramov). According to the World Health Organization, approx. 1.3 billion people (16% of the world population) have a permanent disability, of whom at least 10 million is estimated to be people with amputation; among them, 3 million people hold an upper limb amputation condition (Diego Segura, 2024). The World Health Organization estimates that approximately 1.3 billion people, or 16% of the world's population, have a permanent disability, and among those, it is estimated that 10 million is thought to be people with amputation; that is, 3 million people possess an upper limb amputation condition (Diego Segura, 2024) ,

Overview of BCI

- BCI Revolution and Reach

BCIs translate brain activity into impulses to control external devices, avoiding the traditional neuromuscular pathway. Early applications were for individuals with disabilities, but recently, this evolved into gaming, neuroprosthetics, and virtual reality applications. In nearly all applications, EEG and MEG are primary means of capturing activity in real time within neurons.

Conventional Prosthetics: The original models were mechanical aids-meaning hooks or simple structures to assist movement.

Microprocessor-Controlled Prosthetics: Developed in the 1990s, microprocessors controlled the movements in devices, like the Intelligent Prosthesis. Microprocessors provide a much more natural walk.

AI-Powered Prosthetics: Prosthetics today incorporate artificial intelligence for improving functionality, adaptability, and experience among users.

Electroencephalography (EEG) and Magnetoencephalography (MEG)

- Electroencephalography (EEG):

EEG is the most widely used method within BCI. The possibility of electrical activity recording with high temporal resolution with relatively low cost makes it suitable for real-time monitoring applications, such as motor control and neural feedback. For the same reason, EEG is well-suited for real-time monitoring applications, including motor control and neural feedback. (Kurmanavičiūtė, D., 2024)

Advantages: High temporal resolution, low cost, portability.

Disadvantages: Low spatial resolution, sensitivity to muscle movement-based noise.

- Magnetoencephalography (MEG) Measures:

MEG measures the magnetic fields generated by neural activity and provides higher spatial resolution, which makes it much stronger in localizing the activity of the brain than EEG. It is

best done when complex neural activity is used hence fits very well in cognitive tasks as well as examining other functions of the brain. .(Kurmanavičiūtė, D., 2024)

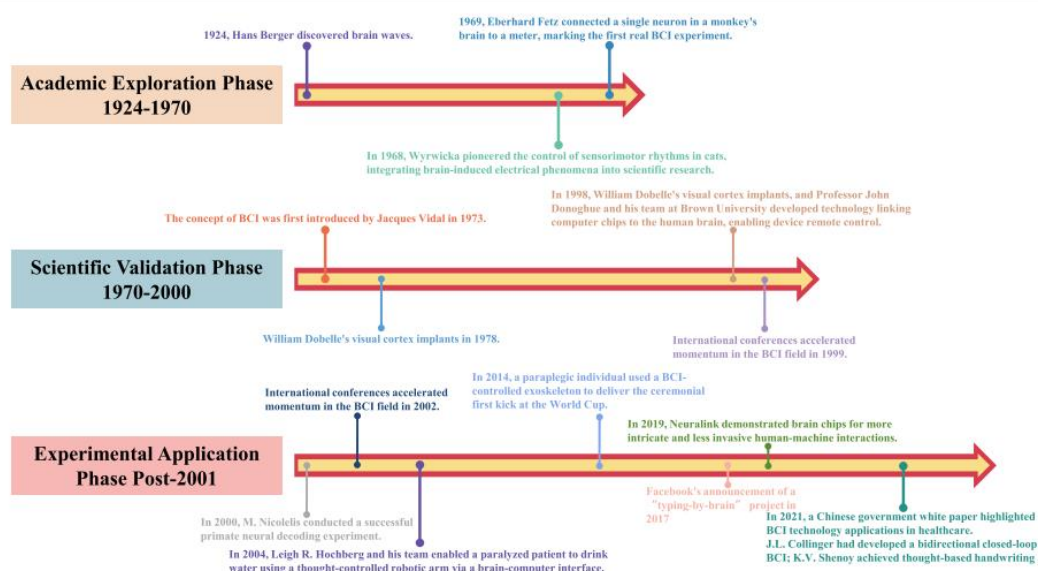
Strength: High Spatial Resolution Non-Invasive Weaknesses: Expensive, requires magnetically shielded rooms as well as cannot be portable. .(Kurmanavičiūtė, D., 2024)

Comparison:- Both are useful in BCIs, but they fulfill the different requirements and serve different purposes depending upon the specific application. The wearable, real-time systems prefer EEG and have more usage, while MEG is more preferred for detailed brain mapping in controlled, clinical settings.

BRAIN-MACHINE AND MUSCLE-MACHINE

As of today, a commonly agreed definition of the term "Brain Robot Interaction" yet to be attained, but it is usually regarded as a derivative field of HRI. Emerging field which helps in establishing a bridge between humans and robots through the translation of brain signals-holds promise in enhancing the daily lives of people with disabilities (Zhang, Y, 2024).

Figure 1, BCI technology has been divided into three groups: invasive, non-invasive, and semi-invasive. Invasive BCIs refer to the electrodes implanted directly within the cerebral cortex, thus involving high signal quality and accurate control but risking surgical interventions as well as biocompatibility issues. On the other hand, non-invasive BCIs utilize external electro-encephalography EEG electrodes to monitor the brain signals with minimal risk, which makes them ideal for initial studies of brain function and clinical diagnosis but with lower quality signal and susceptibility to environmental noise interference. Semi-invasive BCIs place electrodes in subdural or subcortical regions of the brain, thus striking a balance between higher-quality signal compared to non-invasive and lesser risks .compared to invasive methods. (Zhang, Y , 2024) .



Timeline of the development history of brain-computer interfaces. The timeline is divided into three distinct phases: the Academic Exploration Phase (1924–1970), the Scientific Validation Phase (1970–2000), and the Experimental Application Phase (2001–present). This timeline showcases the progression of BCI technology from initial discoveries in brain wave detection to advanced applications in neural control and human-machine interaction, highlighting key milestones and breakthroughs in the field. BCI, brain-computer interface

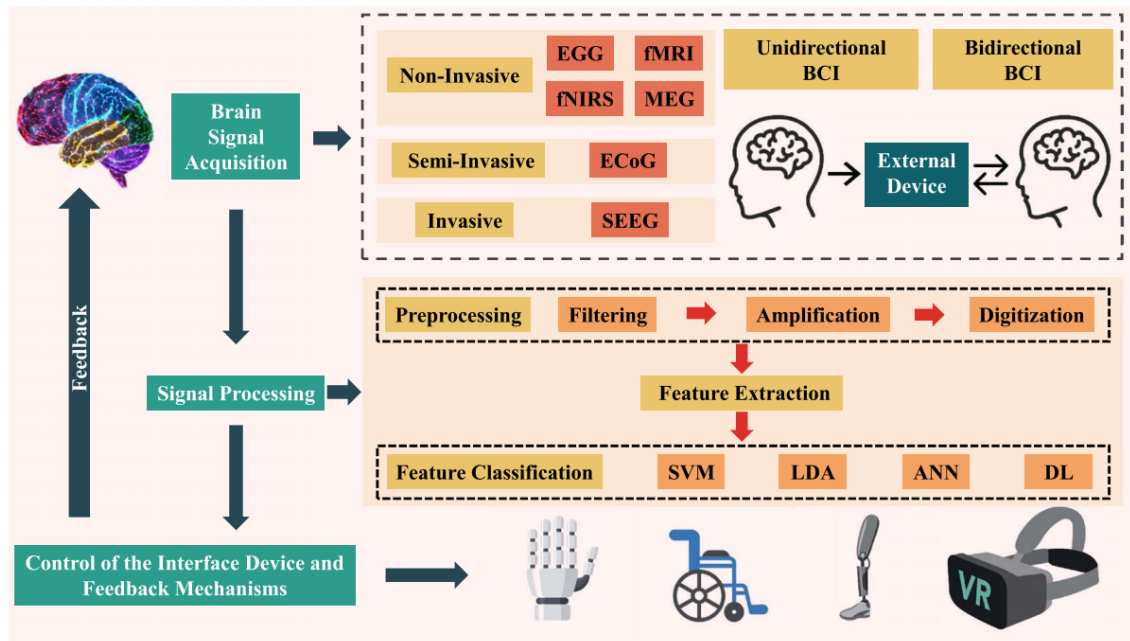


Fig 2-The Classification and Working Principle of Brain-Computer Interfaces. The following figure describes in detail how the process and classification of BCI work in a system-from acquiring brain signals to controlling interfaces and generating feedback. Acquisition of Brain Signals starts from three types in relation to invasion: Non-Invasive includes EEG, fMRI, fNIRS, and MEG, Semi-Invasive involves ECoG, and Invasive involves SEEG. BCIs can be classified as Unidirectional, where the information flows out of the brain to some external device or Bidirectional with information flowing in both directions. The acquired signals are processed using signal processing techniques that include preprocessing that filters, amplifies, and digitizes the signals. Further, feature extraction is performed followed by classification using some algorithms like SVM, LDA, ANN, and DL. With such processed signals, all sorts of interface devices can be controlled, including prosthetic limbs, wheelchairs, and other assistive devices, even virtual reality systems designed to provide feedback to the user, thus closing the loop for BCI. This detailed diagram is, in fact, a whole view of the BCI process, demonstrating all the many ways and technologies in use at every single stage and the complexity and versatility of BCI systems as bridges between neural activity and external device control. ANN, artificial neural network; DL, deep learning; EEG, electroencephalography; ECoG, electrocorticography; fMRI, functional magnetic resonance imaging; fNIRS, functional near-infrared spectroscopy; LDA, linear discriminant analysis; MEG, magnetoencephalography; SEEG, stereoelectroencephalography; SVM, support vector machine; VR, virtual reality

In addition, the BCI can be classified into two categories, which include unidirectional; it sends signals only one way from the brain to a device with prohibition of feedback and adaptation. Conversely, a bidirectional sends feedback that allows adaptation by the brain to improve the amount of control and responses for higher applications.

Prosthetic limbs mimic the movements of a human body with the use of myoelectric signals, produced by electrodes attached to or on residual muscles of an amputated limb. These signals are detected by the electrodes. The system recognizes whether certain muscles have contracted or relaxed or muscle groups produce distinct myoelectric patterns are produced. These patterns are then sensed by the electrodes and interpreted as commands that can control the prosthetic limb.

(Alshamsi H,2016, Md. Moidul Islam1, 2024).

This way, by recognizing the distinct patterns of myoelectric signals, an AI-powered prosthetic can actually mimic the movements the user intends to make. These movements can be adjusted in terms of speed and direction as well as the force involved concerning the detected myoelectric patterns (Md. Moidul Islam1, 2024).

AI in BCI

AI and robotics have transformed technologies, such as prosthetics and exoskeletons, that receive sensory feedback. Techniques like reinforcement learning, neural integration, and machine learning are being applied to enhance the responsiveness and adjustability of prosthetics (Shivani Chopra, 2024).

EEG signals are inherently noisy and complex. AI-based techniques, particularly deep learning, outshine other counterparts in preprocessing EEG to clean up artifacts and make the signal clear.

- The EEG signals are automatically featured by AI techniques like CNNs, extracting patterns associated with brain activity linked to motor imagery
- Traditional Prosthetics: Early designs were characterized by relatively mechanical solutions such as hooks or merely structure that facilitates movement.
- Microprocessor-Controlled Prosthetics: , controlled by microprocessors that enable natural walking movements.
- AI-based Prosthetic Limbs Current prosthetics are designed using artificial intelligence that provides comfort, flexibility, and functionality for the user.

2. Literature Review

The concept of BCIs dates back to the early 20th century, but significant progress has been made in the last few decades, particularly in their application to assistive devices. EEG and MEG are the most commonly used neuroimaging techniques for detecting brain signals in BCI systems. EEG, which records electrical activity on the scalp, is widely used due to its relatively low cost and portability (Kurmanavičiūtė, 2024). However, its spatial resolution is limited, and the signals can be contaminated by muscle activity and environmental noise (Hramov et al., 2021).

MEG, on the other hand, provides superior spatial resolution by measuring the magnetic fields generated by neural activity. It is particularly useful in localizing brain activity but requires expensive equipment and controlled environments, which limits its widespread use (Kurmanavičiūtė, 2024). Both EEG and MEG have been successfully used in controlling robotic prosthetics and exoskeletons, enabling individuals to perform tasks like grasping objects, walking, or interacting with virtual environments (Çevik, 2024).

Recent advancements have incorporated AI to improve the functionality of BCI-controlled devices. AI algorithms, particularly machine learning models, have been used to analyze complex neural signals, filter noise, and predict motor intentions more accurately. These models allow prosthetic devices to adapt to the user's intended movement, enhancing their usability and providing a more natural user experience (Shivani Chopra, 2024). The integration of AI in BCI systems is considered a key area for future development, offering the potential for more intuitive and responsive prosthetics.

HOW PROSTETIC LIMBS WORKS - Signal Processing and Control !

1. Signal Detection

Brain-Computer Interfaces capture the neural activity of the brain, basically carrying user's motor intentions.

- Source of Signals:

BCIs detect signals that emanate from the motor cortex of the human brain, responsible for planning and executing voluntary movements. As a user thinks about moving the hand or fingers, this motored region of the brain fires electrical impulses that can be picked up and interpreted.

- Acquisition Techniques:

Signals are captured depending on the degree of invasiveness;END

EEG: The electrical activity originating in the scalp is captured by attaching electrodes to the scalp. Though non-invasive, EEG signals are usually noisy and include some other activity of the brain or other environmental source.

ECoG: Electrodes inserted on the surface of the brain produce cleaner signals but involve surgery. Intracortical Microelectrodes: These electrodes implanted deep within the brain are known to have the highest resolution and signal fidelity. It poses a great medical risk.

- Detection Problem

Neural signals are weak having a magnitude measured in microvolts and also susceptible to interference. Readings are liable to be distorted by muscle activity, eye blinking, and electrical interference from the environment. Thus, strong preprocessing is necessary.

2. Feature Extraction

Having obtained raw neural signals, one proceeds to extract features specific to motor intentions.

- Signal Preprocessing:

ZERO Noise and artifacts of the raw signal are removed by algorithms, for instance by band-pass filters, to select only relevant frequency bands (e.g., alpha, beta).

- Detecting Relevant Patterns:

Algorithms detect the existence of features in the filtered signals, indicative of the user's intent. Such include,

Frequency Analysis: The variations in the power related to the interesting frequency bands of motor imagery or execution can be detected.

Spatial Filtering: It involves analyzing electrode-specific activity to localize a signal source. The temporal dynamics refer to the dynamics of how signals change over time, for example, the sequence of motor commands.

3. Command Generation

Decoding Intentions:

- The decoded signals are mapped to specific motor actions. For example, a thought that implies closing one's hand activates certain brain areas, which the BCI decodes as a command to close the prosthetic hand.

Action Execution:

- These commands are sent to the actuators in the prosthetic that mimics the movement. For example, motors of a prosthetic hand can change finger position or could be used to apply force to a grip on an object.

Feedback Integration:

- Advanced levels of prosthetic systems have sensory feedback. There is a flow back of information, and that is from touch or grip from the prosthetic to the user. It's a closed-loop system where the user adjusts their action based on the input received in form of tactile.

. Methodology

This paper is based on a **systematic review** of existing literature on the use of EEG and MEG in BCIs, particularly in relation to prosthetic limbs. The research was conducted by analyzing peer-reviewed journal articles, conference proceedings, and patents from reputable sources such as PubMed, IEEE Xplore, and Google Scholar. The primary focus was on articles published from 2010 to 2024, ensuring the inclusion of the most recent advancements in BCI technologies.

The review followed a **qualitative** approach, evaluating the strengths, weaknesses, and applications of EEG and MEG in the context of BCIs. Special attention was given to studies involving AI integration in BCI-controlled prosthetics, as well as the challenges related to signal noise, system latency, and user adaptation.

. Results

Through the review, several key findings were identified:

1. **EEG** remains the most commonly used neuroimaging technique for non-invasive BCIs due to its high temporal resolution and ease of use in real-time applications (Kurmanavičiūtė, 2024). However, its low spatial resolution and susceptibility to external noise limit its effectiveness in certain applications.
2. **MEG**, while offering superior spatial resolution, is less commonly used in prosthetics due to its high cost and the need for specialized environments (Kurmanavičiūtė, 2024). Despite these challenges, MEG has been successfully applied in clinical settings for precise brain mapping and complex cognitive tasks.
3. The **integration of AI** into BCI systems has significantly improved the performance of prosthetic limbs. Machine learning algorithms, such as **Convolutional Neural**

Networks (CNNs), have been used to preprocess EEG data, enhance signal clarity, and predict motor intentions with greater accuracy (Shivani Chopra, 2024).

4. **Signal noise** and **latency** remain significant challenges. EEG signals, in particular, are prone to contamination from muscle activity, eye movement, and environmental interference, which can impact the accuracy of prosthetic control (Çevik, 2024). AI techniques, including deep learning and reinforcement learning, are being explored to mitigate these issues by enhancing signal preprocessing and adapting prosthetic devices to the user's intentions.
 5. Studies have shown that **bidirectional BCIs** (those that allow for feedback from the device to the brain) enhance user control and feedback in prosthetic applications, leading to more intuitive interactions (Zhang, 2024).
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5. Discussion

The results of this review suggest that while EEG and MEG are both valuable for controlling prosthetics, each has distinct advantages and limitations. EEG is more suited for real-time, portable systems, making it ideal for assistive devices used in daily life. However, its limitations in spatial resolution and susceptibility to noise mean that it is less effective for complex tasks requiring high precision. MEG, with its higher spatial resolution, is better suited for controlled clinical environments where detailed brain mapping is necessary but lacks portability.

The integration of AI into BCI systems is a significant step forward. AI algorithms can improve the accuracy of prosthetic limb control by enhancing signal processing and adapting the device's responses based on real-time brain activity. However, challenges such as signal noise and latency remain prevalent. Future research should focus on improving the robustness of signal acquisition techniques and reducing the computational complexity of AI models to facilitate real-time, adaptive control.

Furthermore, **bidirectional BCIs**, which allow for feedback from the prosthetic limb to the brain, hold great promise for creating more responsive and adaptive systems. By providing sensory feedback, these systems can improve user control, leading to more natural interactions with prosthetic devices.

In terms of practical implications, BCIs, particularly when coupled with AI, could vastly improve the quality of life for amputees and individuals with severe physical disabilities. As the technology matures, we can expect more natural, intuitive prosthetic devices that not only replace lost functionality but also enhance the user's ability to interact with their environment.

6. Conclusion

BCIs, powered by EEG and MEG, have shown great potential in improving the functionality of prosthetic limbs. The integration of AI further enhances their capabilities, enabling more responsive, adaptive systems that can better mimic natural human movement. However, challenges such as signal noise, latency, and the need for specialized equipment remain. Moving forward, research should focus on developing more robust signal processing

techniques, enhancing the portability of MEG systems, and exploring bidirectional BCIs for improved feedback and user control.

As these technologies continue to evolve, they promise to significantly improve the lives of individuals with amputations and other disabilities, offering them a higher degree of autonomy and interaction with their surroundings.

6. References

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